

MULTAN INTL, Pakistan

WMO: 416750

Lat: 30.203N

Lon: 71.419E

Elev: 122

StdP: 99.87

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.5	5.9	-1.2	3.5	14.7	0.9	4.1	14.5	5.3	15.9	4.5	16.1	0.1	0	0.399

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	11.0	44.1	23.6	42.6	24.1	41.1	24.7	29.1	36.6	28.7	36.2	28.4	35.9	3.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.5	23.7	32.9	27.0	23.1	32.9	26.5	22.4	33.1	96.6	36.7	94.8	36.2	93.2	36.4	31.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.8	6.8	5.4	2.7	46.7	1.5	1.1	1.6	47.5	0.8	48.1	0.0	48.7	-1.1	49.5		
			1.5	30.4	1.7	0.7	0.3	30.9	-0.7	31.3	-1.6	31.7	-2.8	32.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	26.0	13.1	16.6	22.4	28.8	34.0	35.3	34.3	32.8	31.3	27.2	20.4	14.8	(6)
(7)		DBStd	8.20	2.32	2.72	3.50	3.50	2.76	2.47	2.00	1.65	1.89	2.66	2.29	2.49	(7)
(8)		HDD10.0	6	4	0	0	0	0	0	0	0	0	0	0	2	(8)
(9)		HDD18.3	349	162	61	7	0	0	0	0	0	0	0	6	113	(9)
(10)		CDD10.0	5828	101	184	384	563	743	758	752	707	639	534	313	149	(10)
(11)		CDD18.3	3129	0	12	133	313	484	508	494	449	389	276	68	2	(11)
(12)		CDH23.3	46133	4	120	1330	4203	7814	8466	8034	6883	5591	3077	572	39	(12)
(13)	CDH26.7	29592	0	14	554	2529	5429	6086	5567	4417	3294	1567	134	1	(13)	
(14)	Wind	WSAvg	1.8	0.8	1.3	1.7	2.1	2.4	3.1	3.0	2.9	2.2	1.1	0.6	0.6	(14)
(15)	Precipitation	PrecAvg	181	7	11	17	13	11	15	51	34	13	2	2	6	(15)
(16)		PrecMax	360	43	93	61	66	68	82	231	173	158	23	29	109	(16)
(17)		PrecMin	58	0	0	0	0	0	0	0	0	0	0	0	0	(17)
(18)		PrecStd	75	10	17	18	17	15	23	46	38	26	5	5	17	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.7	28.2	36.4	43.2	46.0	46.2	42.7	40.0	39.2	37.2	31.5	26.2	(19)
(20)			MCWB	14.4	17.4	20.9	21.7	23.7	23.8	27.7	27.4	26.8	23.4	19.5	16.2	(20)
(21)		2%	DB	22.2	26.3	34.0	41.0	44.1	44.3	41.0	38.8	38.0	36.1	29.8	24.5	(21)
(22)			MCWB	13.6	16.7	20.8	21.3	22.8	24.5	27.8	27.5	26.1	23.7	19.2	15.5	(22)
(23)		5%	DB	21.0	24.8	31.9	39.0	42.8	42.7	39.9	37.9	37.0	35.0	28.2	23.0	(23)
(24)			MCWB	13.3	16.1	20.0	21.1	22.9	25.3	27.8	27.5	25.9	23.0	18.3	14.7	(24)
(25)		10%	DB	19.5	23.0	29.9	36.8	41.1	41.1	38.8	36.9	36.0	33.5	26.8	21.4	(25)
(26)			MCWB	12.5	15.1	19.4	20.8	22.8	25.9	27.8	27.6	26.0	22.5	17.9	13.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.8	19.4	23.2	23.8	27.2	29.0	29.8	29.6	29.0	26.2	21.6	17.8	(27)
(28)			MCD B	21.2	25.0	33.0	38.0	38.3	38.5	37.6	35.5	35.6	33.7	28.1	23.7	(28)
(29)		2%	WB	14.8	18.2	22.1	22.9	26.2	28.5	29.2	29.0	28.4	25.5	20.7	16.6	(29)
(30)			MCD B	20.2	24.3	31.1	36.2	37.9	38.4	37.1	35.5	34.8	32.9	27.0	22.3	(30)
(31)		5%	WB	14.1	17.2	21.0	22.2	25.3	28.0	28.8	28.6	27.8	24.8	19.8	15.6	(31)
(32)			MCD B	19.1	23.2	29.2	34.7	37.6	38.3	36.7	35.1	34.0	32.0	25.6	21.1	(32)
(33)		10%	WB	13.3	16.2	20.1	21.5	24.2	27.5	28.5	28.3	27.2	23.9	19.0	14.6	(33)
(34)			MCD B	18.0	21.4	28.1	33.6	37.4	37.4	36.4	34.7	33.6	31.1	24.9	19.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.9	12.1	12.3	13.8	13.2	11.0	8.5	8.0	9.2	12.2	13.0	12.9	(35)
(36)			MCDBR	14.3	14.2	14.3	15.9	15.2	13.4	9.4	8.9	10.2	12.5	14.3	14.9	(36)
(37)			MCWBR	7.0	6.0	4.9	4.3	4.2	3.9	2.8	2.8	3.2	3.9	5.4	7.2	(37)
(38)		5% WB	MCD BR	12.2	12.3	13.4	14.3	12.5	9.7	8.5	8.1	8.7	10.7	12.0	13.0	(38)
(39)			MCWBR	6.5	5.6	5.0	4.5	4.8	3.9	3.1	3.0	3.4	3.7	5.4	6.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.502	0.504	0.526	0.595	0.648	0.756	0.818	0.732	0.624	0.622	0.616	0.553	(40)	
(41)		taud	1.819	1.868	1.847	1.735	1.650	1.509	1.441	1.578	1.731	1.646	1.619	1.725	(41)	
(42)		Ebn at Noon	691	745	764	729	693	619	579	624	675	630	579	613	(42)	
(43)		Edh at Noon	178	184	201	232	254	292	311	269	223	228	216	187	(43)	
(44)	All-Sky Solar Radiation	RadAvg	3.17	4.17	5.43	6.50	7.02	6.73	6.00	5.82	5.56	4.78	3.64	3.15	(44)	
(45)		RadStd	0.26	0.31	0.29	0.23	0.19	0.27	0.30	0.29	0.28	0.20	0.23	0.16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page